

Conditional Probability and Bayes' Rule

Conditional questions, denominator audits, Bayesian updates, forecasts, and uncertainty controls.

Library of the Quant

Educational reference manual

Manual Profile

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Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

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How to use this manual

INTERMEDIATE

1. WRITE THE CONDITIONING BAR

Translate every claim into $P(A|B)$, $P(B|A)$, or a conditional distribution. Name the observation unit, horizon, event boundaries, and eligibility population before calculating anything.

2. AUDIT THE DENOMINATOR

Conditional probabilities reuse the same overlap with different reference populations. Read raw counts before percentages and check missing, unknown, rejected, or censored observations.

3. SEPARATE PRIOR FROM EVIDENCE

Bayes' rule starts from a base rate and updates it using the likelihood of observed evidence under competing hypotheses. A compelling signal can still imply a modest posterior when the event is rare.

4. CARRY UNCERTAINTY

Report sample size, intervals, posterior spread, and support. Sparse conditional cells and aggressive priors can both create false certainty.

5. VALIDATE POINT IN TIME

Every conditioning variable and piece of evidence must have been available at the decision cutoff. Replay feature values, event timestamps, priors, and sequential state exactly as production would have seen them.

6. TEST PROBABILITY MEANING

Use reliability curves, confusion tables, posterior predictive checks, and regime slices. Ranking quality alone does not justify sizing decisions from probability values.

DECISION STANDARD

Promote a conditional estimate only when its population, timing, evidence, assumptions, uncertainty, calibration, and consuming decision are all documented and reproducible.

Conditional probability notation

INTERMEDIATE

SYMBOL	TYPE	MEANING
Omega	sample space	All eligible outcomes
A, B, E	events / evidence	Sets of outcomes
H	hypothesis	State or model being updated
A and B	intersection	Both events occur
A or B	union	At least one occurs
$P(A B)$	conditional probability	Chance of A inside B
$P(A,B)$	joint probability	Chance of A and B
$P(A)$	marginal / prior	Chance before current evidence
$P(E H)$	likelihood term	Evidence chance under H
$P(H E)$	posterior	Updated chance of H

DIRECTION RULE

The vertical bar is directional. $P(A|B)$ uses B as its denominator, while $P(B|A)$ uses A; Bayes' rule is the justified conversion between them, not permission to swap them.

OPERATIONAL CONTRACT

Attach grain, horizon, event clock, cutoff, population, missing policy, count, weighting, and estimator to every probability. These fields are part of the number, not optional metadata.

Conditional inference workflow

INTERMEDIATE

1

Define the decision

Action, payoff, horizon, and target population

2

Specify events

Grain, boundaries, cutoff, and eligibility

3

Measure base rates

Prior prevalence with uncertainty and lineage

4

Model evidence

Joint cells, likelihoods, and dependence

5

Update

Normalize posterior or estimate conditional law

6

Diagnose

Counts, support, calibration, and sensitivity

7

Validate in time

Replay, holdout, regimes, and selection controls

8

Publish safely

Artifact IDs, health gates, fallback, rollback

HARD GATE

No conditional or posterior probability reaches a capital decision until its denominator, evidence timing, dependence assumptions, uncertainty, and calibration can be replayed.

Conditional probability as a restricted world

INTERMEDIATE

ONE-SENTENCE MEANING

Conditional probability measures an event after the sample space has been restricted to cases where the conditioning event occurred. The denominator changes, so the answer is about a new reference population rather than a revised label on the old one.

MEMORY JOGGER

Read $P(A|B)$ as the fraction of B cases that also belong to A. First enter the B world, discard everything outside it, and only then count A.

WHY IT MATTERS IN QUANT RESEARCH

Signals, regimes, fills, defaults, and event reactions are conditional questions because the relevant chance depends on information available at decision time. Explicit conditioning prevents an unconditional average from being mistaken for a live forecast.

FORMULA INTUITION

$P(A|B) = P(A \text{ and } B) / P(B)$, for $P(B) > 0$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Suppose 200 trading days include 50 high-volatility days and 18 of those close below -1%. Then $P(\text{loss below } -1\% \mid \text{high volatility}) = 18/50 = 36\%$, not $18/200 = 9\%$.

PYTHON / SYSTEM IMPLEMENTATION

Store numerator count, denominator count, event definitions, eligibility rules, and cutoff time together. Unit tests should reconstruct the conditional rate from the underlying event flags and reject zero-denominator groups.

CONFUSIONS TO AVOID

Reversing the bar changes the question: $P(A|B)$ generally differs from $P(B|A)$. Conditioning on a post-outcome variable can also create leakage even when the arithmetic is correct.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature and event stores define point-in-time event flags; the probability service publishes conditional estimates with coverage and uncertainty; decision rules consume them only for matching populations.

RELATED CONCEPTS AND QUICK RECALL

Related: joint probability, reference class, event filter. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Sample spaces, events, and measurable questions

ONE-SENTENCE MEANING

A sample space lists possible outcomes, while an event is a set of outcomes whose probability can be evaluated. Conditioning is reliable only when outcomes, event boundaries, and observation timing are defined before counting.

MEMORY JOGGER

A probability question needs a universe and a yes-or-no membership rule. If two researchers cannot classify the same record identically, the event is not operationally defined.

WHY IT MATTERS IN QUANT RESEARCH

Quant research often mixes bars, orders, instruments, and days in one analysis. A precise sample space prevents denominator drift, duplicate exposure, and comparisons between incompatible units of observation.

FORMULA INTUITION

A is a subset of Ω ; $P(\Omega)=1$ and $0 \leq P(A) \leq 1$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For an order-level fill model, outcomes are submitted child orders, not market days. The event A might be 'filled within 30 seconds,' with cancellations and rejected orders handled by a declared eligibility policy.

PYTHON / SYSTEM IMPLEMENTATION

Create an event contract containing grain, entity key, clock, inclusion rules, event expression, data version, and owner. Generate stable event IDs so backtests and production monitors refer to the same definition.

CONFUSIONS TO AVOID

Changing the unit from orders to shares weights large orders more heavily and answers a different question. Silent changes to universe membership can move a conditional probability without any economic change.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The semantic layer owns event contracts and exposes reusable flags. Downstream notebooks query those flags instead of reimplementing business logic in ad hoc filters.

RELATED CONCEPTS AND QUICK RECALL

Related: sample space, event algebra, observation grain. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Joint probability and intersections

ONE-SENTENCE MEANING

Joint probability is the chance that multiple events occur together. It is the bridge between marginal frequencies and conditional statements because $P(A \text{ and } B)$ appears in both conditional-probability formulas.

MEMORY JOGGER

The intersection is overlap, not addition. Count rows satisfying every condition and keep the shared observation unit fixed.

WHY IT MATTERS IN QUANT RESEARCH

Joint events describe combinations such as a momentum signal and a volatility regime, an order fill and adverse price move, or a loss and a liquidity shock. They reveal whether conditions coincide often enough to support a segmented estimate.

FORMULA INTUITION

$P(A \text{ and } B) = P(A|B)P(B) = P(B|A)P(A)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If 30% of days are risk-off and 40% of risk-off days produce a negative momentum signal, the joint chance is $0.40 \times 0.30 = 12\%$. This is the frequency of days satisfying both conditions.

PYTHON / SYSTEM IMPLEMENTATION

Build joint flags with boolean AND, then reconcile $\text{count}(A \text{ and } B)$ from both source events. For weighted data, document weights and verify the joint mass never exceeds either marginal mass.

CONFUSIONS TO AVOID

Adding $P(A)$ and $P(B)$ double-counts their overlap unless inclusion-exclusion is used. Pairwise joint frequencies also do not fully specify dependence among three or more events.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The cohort engine materializes joint states used by model training, scenario analysis, and monitoring. Coverage thresholds stop sparse combinations from being promoted as stable signals.

RELATED CONCEPTS AND QUICK RECALL

Related: intersection, co-occurrence, chain rule. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Marginal probability and aggregation

INTERMEDIATE

ONE-SENTENCE MEANING

A marginal probability describes an event after summing or integrating over other variables. It is an average across hidden states, weighted by how frequently those states occur.

MEMORY JOGGER

Marginalize means remove a dimension by adding its probability mass. The result can change because state frequencies change even when every within-state probability stays fixed.

WHY IT MATTERS IN QUANT RESEARCH

Aggregate hit rates, fill rates, and default rates are marginal quantities. Researchers must decompose them before attributing movement to model quality rather than a different mix of regimes, instruments, or clients.

FORMULA INTUITION

$P(A) = \sum_b P(A|B=b)P(B=b)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A signal wins 55% in calm markets and 40% in stress. If the market mix shifts from 80/20 to 40/60, the marginal win rate falls from 52% to 46% with no change inside either regime.

PYTHON / SYSTEM IMPLEMENTATION

Persist both conditional cells and their population weights. A reconciliation query should reproduce the aggregate from the weighted cells and flag missing or overlapping states.

CONFUSIONS TO AVOID

An aggregate rate is not automatically transportable to a population with different state weights. Equal-weighting group rates estimates a different target than weighting by actual frequency.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metrics layer publishes aggregate results with a mix-shift decomposition. Monitoring distinguishes changes in conditional behavior from changes in population composition.

RELATED CONCEPTS AND QUICK RECALL

Related: law of total probability, mixtures, standardization. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Complements, unions, and inclusion-exclusion

ONE-SENTENCE MEANING

A complement contains outcomes outside an event, while a union contains outcomes in either event or both. Inclusion-exclusion subtracts the overlap that would otherwise be counted twice.

MEMORY JOGGER

Not A uses one minus; A or B uses add then subtract overlap. Translate ordinary language carefully because 'or' in probability usually includes both.

WHY IT MATTERS IN QUANT RESEARCH

Risk limits often combine failure events, such as a loss, stale quote, or liquidity breach. Correct union probabilities prevent multiple simultaneous alerts from inflating the estimated chance of any failure.

FORMULA INTUITION

$P(\text{not } A) = 1 - P(A)$; $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If $P(\text{gap loss}) = 4\%$, $P(\text{liquidity breach}) = 7\%$, and both occur on 2% of days, the chance of at least one is $4\% + 7\% - 2\% = 9\%$. Adding marginals alone would report 11%.

PYTHON / SYSTEM IMPLEMENTATION

Implement event algebra on immutable booleans and compare observed union counts with the formula. For many events, also estimate the union directly because high-order intersections become cumbersome.

CONFUSIONS TO AVOID

Assuming disjointness because event names differ is unsafe. The complement of 'return above zero' also includes zero unless the event boundary explicitly assigns it elsewhere.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The control framework composes atomic breach flags into policy-level alerts. The lineage graph preserves which overlaps were included so audit results are reproducible.

RELATED CONCEPTS AND QUICK RECALL

Related: complement rule, union bound, overlap. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

The multiplication rule and probability chains

INTERMEDIATE

ONE-SENTENCE MEANING

The multiplication rule factors a joint probability into a sequence of conditional probabilities. Any ordering is mathematically valid, but a useful order follows the mechanism and available information.

MEMORY JOGGER

Build a path by multiplying branch chances. Each branch probability is conditioned on everything that happened earlier on that path.

WHY IT MATTERS IN QUANT RESEARCH

Multi-stage trading events - signal generation, order submission, fill, favorable move, and exit - naturally form a chain. Stage-level probabilities reveal where an apparent edge is lost operationally.

FORMULA INTUITION

$P(A_1, \dots, A_k) = P(A_1) \text{ product from } i=2..k \text{ of } P(A_i | A_1, \dots, A_{i-1})$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A setup appears on 10% of days, 70% pass risk checks, 80% fill, and 55% of fills win. Under these conditional rates, the full winning-path chance is $0.10 \times 0.70 \times 0.80 \times 0.55 = 3.08\%$.

PYTHON / SYSTEM IMPLEMENTATION

Represent stages as a directed event graph with counts at each edge. Validate that child counts do not exceed parent counts and report uncertainty where a branch becomes sparse.

CONFUSIONS TO AVOID

Multiplying unconditional stage rates is wrong when stage populations differ. A chain can also hide selection effects if downstream stages depend on unrecorded decisions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The execution funnel records every transition from candidate to outcome. Researchers can separate signal weakness from risk rejection, non-fill, or exit-quality problems.

RELATED CONCEPTS AND QUICK RECALL

Related: joint distribution, probability tree, funnel analysis. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Law of total probability

ONE-SENTENCE MEANING

The law of total probability reconstructs an event probability by summing its conditional probability across a complete, non-overlapping partition. It formalizes how state-specific behavior and state frequency combine.

MEMORY JOGGER

Split the world into exhaustive states, solve inside each state, weight by state probability, then add. Every outcome must belong to exactly one state.

WHY IT MATTERS IN QUANT RESEARCH

Regime models, venue routing, credit grades, and volatility buckets use this decomposition. It supports scenario analysis by changing either state probabilities or within-state outcomes separately.

FORMULA INTUITION

$P(A) = \sum_i P(A|B_i)P(B_i)$, where B_i form a partition. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If fills occur with probabilities 90%, 70%, and 45% in liquid, normal, and stressed states weighted 20%, 60%, and 20%, total fill probability is $18\% + 42\% + 9\% = 69\%$.

PYTHON / SYSTEM IMPLEMENTATION

Check that state flags are mutually exclusive, exhaustive, and point-in-time. Publish cell contributions $P(A|B_i)P(B_i)$ so users can see which regimes drive the total.

CONFUSIONS TO AVOID

Overlapping labels double-count mass, while an 'unknown' state omitted from the partition loses mass. Using future realized volatility to define states introduces hindsight.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The scenario service stores state weights and conditional response curves separately. Stress tests can replace the weights without refitting the conditional mechanics.

RELATED CONCEPTS AND QUICK RECALL

Related: partitions, marginalization, mixture models. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Independence versus conditional independence

ONE-SENTENCE MEANING

Events are independent when knowing one does not change the probability of the other. Conditional independence is narrower: dependence may disappear after a specified common state is known.

MEMORY JOGGER

Independence means information adds nothing for that probability question. Always name the conditioning set because independence can appear or disappear after stratification.

WHY IT MATTERS IN QUANT RESEARCH

Feature assumptions, naive Bayes models, risk aggregation, and simulation engines often rely on independence. Violations can understate joint tail risk or double-count evidence from correlated signals.

FORMULA INTUITION

A independent B iff $P(A \text{ and } B) = P(A)P(B)$; A independent B | C iff $P(A, B | C) = P(A | C)P(B | C)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Stock and bond losses may be positively associated overall because both respond to inflation shocks. Within a fixed macro regime they may be much closer to independent, although that must be tested rather than assumed.

PYTHON / SYSTEM IMPLEMENTATION

Compare observed and factorized joint frequencies with confidence intervals, then repeat across relevant states. For models, monitor residual dependence after conditioning and use clustered resampling.

CONFUSIONS TO AVOID

Zero correlation does not imply independence except under special distributional conditions. Pairwise independence also does not guarantee mutual independence among several variables.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The dependency registry records which models assume conditional independence and which diagnostics support it. Stress tests deliberately break the assumption to measure downstream sensitivity.

RELATED CONCEPTS AND QUICK RECALL

Related: dependence, graphical models, mutual information. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Base rates and reference classes

INTERMEDIATE

ONE-SENTENCE MEANING

A base rate is the prior frequency of an outcome in a defined reference population. Conditional evidence should update that rate, not erase it.

MEMORY JOGGER

Start with how common the event is before seeing the signal. Rare outcomes remain rare unless the evidence has a sufficiently strong likelihood ratio.

WHY IT MATTERS IN QUANT RESEARCH

Fraud, default, crashes, and large alpha opportunities are low-base-rate events. Ignoring prevalence creates impressive-looking hit rates that deliver many false positives in production.

FORMULA INTUITION

Prior odds = $P(A)/(1-P(A))$; posterior odds = prior odds x likelihood ratio. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A crash alert has 80% sensitivity and a 10% false-positive rate, while crashes occur on 1% of days. Among alerts, crash probability is only 7.5%, because ordinary days dominate the population.

PYTHON / SYSTEM IMPLEMENTATION

Estimate base rates on the deployment population and horizon, with an as-of date and uncertainty range. Reweight training data back to deployment prevalence before interpreting predicted probabilities.

CONFUSIONS TO AVOID

A balanced machine-learning sample deliberately changes prevalence and cannot supply calibrated production probabilities directly. The narrowest convenient subgroup may also be too sparse to be a stable reference class.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model registry stores training prevalence, deployment prevalence, and any correction. Alert dashboards show positive predictive value alongside sensitivity and false-positive rate.

RELATED CONCEPTS AND QUICK RECALL

Related: prevalence, prior probability, rare-event classification. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Two-way tables and denominator discipline

INTERMEDIATE

ONE-SENTENCE MEANING

A two-way table displays joint counts for two categorical events and exposes every denominator used by conditional probabilities. It is the fastest audit tool for reversing conditions or hiding class imbalance.

MEMORY JOGGER

Rows and columns are populations; cells are overlaps. Divide a cell by its row total for one condition and by its column total for the reverse condition.

WHY IT MATTERS IN QUANT RESEARCH

Contingency tables clarify signal precision, recall, fill behavior, and risk-event co-occurrence. Raw counts beside rates make sparse estimates and prevalence visible.

FORMULA INTUITION

$P(A|B) = n(A \text{ and } B) / n(B)$; $P(B|A) = n(A \text{ and } B) / n(A)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

With 8 true alerts, 92 false alerts, 2 missed events, and 898 quiet non-alerts, $P(\text{event}|\text{alert}) = 8\%$, while $P(\text{alert}|\text{event}) = 80\%$. Both use the same true-positive cell but different denominators.

PYTHON / SYSTEM IMPLEMENTATION

Generate counts before percentages, include row and column totals, and assert the grand total matches the eligible population. Add minimum-count warnings and exact intervals for rare cells.

CONFUSIONS TO AVOID

Reporting only normalized percentages can conceal tiny cell counts. Excluding 'unknown' outcomes after seeing their label changes the denominator and may bias both directions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evaluation service emits a canonical confusion table for each cutoff, horizon, and cohort. Dashboards link every displayed rate back to its count table.

RELATED CONCEPTS AND QUICK RECALL

Related: confusion matrix, precision, recall. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Probability trees and path accounting

INTERMEDIATE

ONE-SENTENCE MEANING

A probability tree represents sequential uncertainty with conditional branches whose probabilities sum to one at each node. Path probabilities multiply, and terminal paths add when they lead to the same final event.

MEMORY JOGGER

Multiply down a branch and add across alternative branches. Label every split with what is known at that point in time.

WHY IT MATTERS IN QUANT RESEARCH

Trees make execution, event, and scenario models auditable because they expose missing branches and conditional dependencies. They are especially useful when a rare path carries disproportionate P&L.

FORMULA INTUITION

$P(\text{path}) = \text{product of branch probabilities}$; $P(\text{outcome}) = \text{sum of relevant path probabilities}$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A news event is positive with probability 30%. The strategy wins 70% after positive news and 40% otherwise, so total win probability is $0.30 \times 0.70 + 0.70 \times 0.40 = 49\%$.

PYTHON / SYSTEM IMPLEMENTATION

Store nodes, parent IDs, conditions, branch probabilities, and terminal payoffs in a table. Validate local branch sums, compute path mass recursively, and reconcile terminal mass to one.

CONFUSIONS TO AVOID

A branch probability copied from a different parent population breaks the tree. Recombining paths that share state may require a graph or dynamic model rather than duplicating inconsistent nodes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The scenario engine serializes trees as versioned artifacts used by stress tests and decision explanations. Observed outcomes later update branch calibration by node.

RELATED CONCEPTS AND QUICK RECALL

Related: chain rule, scenario tree, decision tree. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Conditional distributions, not just rates

INTERMEDIATE

ONE-SENTENCE MEANING

A conditional distribution describes the full range of an outcome given information, not merely the chance of one event. Its center, spread, asymmetry, and tails may all change with the condition.

MEMORY JOGGER

Conditioning reshapes an entire distribution. A stable conditional mean can coexist with much wider risk or a new downside tail.

WHY IT MATTERS IN QUANT RESEARCH

Trading decisions depend on payoff distributions because sizing and risk cannot be recovered from a hit rate alone. Regime-conditional distributions support expected shortfall, scenario loss, and utility calculations.

FORMULA INTUITION

$f(x|b)=f(x,b)/P(B=b)$ for discrete B ; conditional CDF $F(x|b)=P(X\leq x|B=b)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A strategy averages 5 bps in both calm and stressed markets, but stressed returns have five times the volatility and a -200 bp tail. The same mean implies a very different position size.

PYTHON / SYSTEM IMPLEMENTATION

Estimate conditional quantiles, ECDFs, sample size, and tail coverage for every material cohort. Use partial pooling or hierarchical models when direct cells are too sparse.

CONFUSIONS TO AVOID

Comparing only conditional means can miss changing risk. Arbitrary binning can create unstable distributions and discontinuities near bucket boundaries.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The forecast API returns a distribution object with quantiles, moments, scenario samples, and calibration metadata. The allocator consumes the distribution rather than a single expected value.

RELATED CONCEPTS AND QUICK RECALL

Related: conditional density, quantile, heteroskedasticity. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Conditioning on continuous variables

INTERMEDIATE

ONE-SENTENCE MEANING

When the condition is continuous, an exact value usually has zero probability, so conditional behavior is represented by a function, density, or local neighborhood. The estimation method determines how information is shared across nearby values.

MEMORY JOGGER

Do not filter for an exact floating-point value and call the result conditional probability. Learn a smooth or structured relationship across the conditioning variable.

WHY IT MATTERS IN QUANT RESEARCH

Volatility, spread, imbalance, and signal strength are continuous conditioning variables. Their response curves help choose thresholds and identify nonlinear regions where behavior changes.

FORMULA INTUITION

$m(z) = P(A|Z=z)$ or $E[Y|Z=z]$, estimated by a model or smoother. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Fill probability may decline smoothly from 90% at a one-tick spread to 35% at ten ticks. A logistic model can estimate the curve without creating nine brittle buckets.

PYTHON / SYSTEM IMPLEMENTATION

Use splines, generalized linear models, isotonic regression, or carefully cross-validated bins. Plot support density and uncertainty so extrapolated regions are visually distinct.

CONFUSIONS TO AVOID

A flexible curve can memorize noise where data are thin. Conditioning on a transformed variable requires preserving the transform version and its point-in-time inputs.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature service publishes continuous variables; the conditional-response model records support, monotonic constraints, and calibration by range. Runtime requests outside support trigger a fallback.

RELATED CONCEPTS AND QUICK RECALL

Related: regression function, smoothing, support. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Conditional expectation and the information set

INTERMEDIATE

ONE-SENTENCE MEANING

Conditional expectation is the probability-weighted mean after accounting for available information. It is a random variable before the information is observed and a numerical forecast afterward.

MEMORY JOGGER

The unconditional mean is a baseline; the conditional mean is what the model knows now. Every forecast should name the information set that makes it conditional.

WHY IT MATTERS IN QUANT RESEARCH

Alpha models estimate expected future return given current features, while cost models estimate expected implementation cost given an order and market state. Both are conditional expectations with strict timing requirements.

FORMULA INTUITION

$m(X_t) = E[R_{t+h} | \text{information available at } t]$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

The unconditional next-day return is 1 bp, but the model estimates 9 bps when trend is strong and volatility is moderate. That 9 bp forecast is meaningful only for matching states and a fixed horizon.

PYTHON / SYSTEM IMPLEMENTATION

Timestamp each feature by availability, assemble training rows at the decision clock, and replay the exact feature view historically. Compare conditional forecasts against the unconditional baseline out of sample.

CONFUSIONS TO AVOID

Using revised economic data or end-of-day values in an intraday information set creates leakage. A complex forecast can also have worse decision value than the unconditional mean after costs.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The point-in-time feature store defines the information set, the forecast service emits conditional means, and monitoring checks calibration and economic value by forecast bucket.

RELATED CONCEPTS AND QUICK RECALL

Related: regression, filtration, forecast calibration. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Conditional variance and heteroskedastic risk

INTERMEDIATE

ONE-SENTENCE MEANING

Conditional variance measures uncertainty remaining after current information is used. It can change over time even when unconditional variance is stable because volatility clusters and regimes persist.

MEMORY JOGGER

The mean asks where; conditional variance asks how uncertain, given what is known now. Risk forecasts should move when the information set indicates changing dispersion.

WHY IT MATTERS IN QUANT RESEARCH

Position sizing, option valuation, margin, and stop design require state-dependent risk. Treating variance as constant often overallocates during stress and underuses capital during quiet periods.

FORMULA INTUITION

$\text{Var}(Y|X) = E[(Y - E[Y|X])^2 | X]$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A strategy has 1% daily volatility overall, but forecasts 0.6% in calm markets and 2.2% after a volatility shock. The same expected return produces sharply different risk-adjusted attractiveness.

PYTHON / SYSTEM IMPLEMENTATION

Model squared or absolute residuals with lagged information, then validate interval coverage by forecast bucket. Separate mean-model error from variance-model error and monitor both.

CONFUSIONS TO AVOID

Using realized future volatility as a contemporaneous condition is leakage. Squared returns are noisy proxies, so apparent precision in a short sample is suspect.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk service consumes conditional variance and covariance forecasts at the allocation timestamp. Breach monitoring compares predicted intervals with realized outcomes.

RELATED CONCEPTS AND QUICK RECALL

Related: ARCH effects, volatility clustering, prediction intervals. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Iterated expectations and nested information

INTERMEDIATE

ONE-SENTENCE MEANING

The law of iterated expectations says an unconditional mean equals the average of conditional means, and a coarser forecast equals the average of a finer forecast given the coarser information. It provides a reconciliation rule for nested models.

MEMORY JOGGER

Forecast with detailed information, then average away detail to recover the broader forecast. If reconciliation fails, samples, weights, or definitions differ.

WHY IT MATTERS IN QUANT RESEARCH

Desk, sector, and portfolio forecasts should aggregate coherently. The tower property also separates within-state performance from changes in state frequency.

FORMULA INTUITION

$E[Y] = E[E[Y|X]]$; if G is contained in H , $E[E[Y|H]|G] = E[Y|G]$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Calm and stress states occur 75% and 25% of the time with expected returns 4 and -8 bps. The unconditional expectation is 1 bp, matching the weighted conditional means.

PYTHON / SYSTEM IMPLEMENTATION

Create reconciliation tests using identical rows and weights at each aggregation level. Store contribution tables so failures can be traced to coverage, missing values, or inconsistent taxonomy.

CONFUSIONS TO AVOID

Averaging subgroup forecasts equally violates the identity when groups have different mass. Overlapping cohorts cannot be treated as a partition without explicit allocation weights.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The forecast hierarchy validates strategy-to-book-to-portfolio consistency before publication. Attribution reports separate mix effects from within-state forecast changes.

RELATED CONCEPTS AND QUICK RECALL

Related: tower property, hierarchy, aggregation. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Selection bias and collider conditioning

INTERMEDIATE

ONE-SENTENCE MEANING

Conditioning on a variable caused by two other variables can create an association between them even when none existed in the source population. The selected sample answers a different causal question from the full population.

MEMORY JOGGER

If two paths both affect admission into the dataset, analyzing only admitted cases can make the paths look related. Selection is part of the data-generating process, not a harmless filter.

WHY IT MATTERS IN QUANT RESEARCH

Filled orders, surviving funds, listed securities, and approved trades are selected populations. Studying outcomes only after selection can misattribute execution, strategy, or risk behavior.

FORMULA INTUITION

If $X \rightarrow S \leftarrow Y$, conditioning on collider S can induce dependence between X and Y . Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Order urgency and market liquidity both increase fill probability. Among filled orders only, urgent orders may appear concentrated in poor liquidity even if urgency and liquidity were independent at submission.

PYTHON / SYSTEM IMPLEMENTATION

Draw a causal graph, log the selection decision, and retain rejected or censored cases when possible. Use inverse-probability weighting or sensitivity analysis only when assumptions are explicit and testable.

CONFUSIONS TO AVOID

Adding more controls is not automatically safer; controlling for a collider can increase bias. A predictive association inside a selected sample may not transport to the decision population.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The research dataset registry records selection stages and coverage. Model cards state the target population and warn when production traffic differs from the selected training cohort.

RELATED CONCEPTS AND QUICK RECALL

Related: sample selection, survivorship bias, causal graphs. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Simpson's paradox and aggregation reversals

INTERMEDIATE

ONE-SENTENCE MEANING

Simpson's paradox occurs when an aggregate association reverses or disappears after conditioning on a relevant grouping variable. Unequal group sizes and different baseline rates drive the reversal.

MEMORY JOGGER

The total can tell the opposite story from every subgroup. Before acting on an aggregate comparison, inspect the strata that determine the population mix.

WHY IT MATTERS IN QUANT RESEARCH

Strategy win rates, execution quality, and model accuracy can move because instrument or regime composition changes. A pooled metric may reward a system that receives easier cases.

FORMULA INTUITION

$P(A|B)$ comparisons can reverse after conditioning on C when $P(C|B)$ differs. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Router A fills 80% in liquid names and 40% in illiquid names; Router B fills 75% and 35%. If A receives mostly illiquid orders and B mostly liquid orders, B can lead in aggregate despite losing in both groups.

PYTHON / SYSTEM IMPLEMENTATION

Publish stratified rates, group weights, and a standardized comparison using common weights. Predefine material strata and avoid searching many partitions for a favorable reversal.

CONFUSIONS TO AVOID

A reversal does not prove which view is causally correct. Conditioning on a mediator or collider can create a misleading stratified result, so the causal role of the grouping variable matters.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experimentation platform reports pooled, stratified, and mix-adjusted effects. Reviewers must approve the target population and standardization weights before launch decisions.

RELATED CONCEPTS AND QUICK RECALL

Related: confounding, standardization, mix shift. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Bayes' rule as reverse conditioning

INTERMEDIATE

ONE-SENTENCE MEANING

Bayes' rule converts the probability of evidence given a hypothesis into the probability of the hypothesis given the evidence. It combines a prior base rate with how diagnostic the evidence is.

MEMORY JOGGER

Prior times likelihood, then normalize. The rule does not create information; it reweights hypotheses according to how well each predicts what was observed.

WHY IT MATTERS IN QUANT RESEARCH

Researchers use Bayes' rule for regime inference, event classification, model uncertainty, and sequential belief updates. It prevents a persuasive signal from overwhelming a very low base rate without sufficient evidence.

FORMULA INTUITION

$P(H|E) = P(E|H)P(H)/P(E)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A regime model assigns 20% prior probability to stress. A spread widening is twice as likely in stress as calm, so posterior odds double from 1:4 to 1:2, giving posterior stress probability 33.3%.

PYTHON / SYSTEM IMPLEMENTATION

Compute unnormalized weights for every hypothesis, sum them for the evidence probability, and normalize. Use log probabilities for long sequences and test that posterior mass sums to one.

CONFUSIONS TO AVOID

$P(E|H)$ is not $P(H|E)$, and a high likelihood under H does not make H probable if alternatives predict E equally well. Priors and hypothesis coverage must be declared.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The inference service consumes prior state probabilities and observation models, then publishes posterior state weights with evidence lineage. Allocation uses those weights in a mixture forecast.

RELATED CONCEPTS AND QUICK RECALL

Related: inverse probability, normalization, base rate. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Odds form and likelihood ratios

INTERMEDIATE

ONE-SENTENCE MEANING

The odds form of Bayes' rule updates prior odds by multiplying by a likelihood ratio. It isolates the evidence strength from the starting prevalence.

MEMORY JOGGER

Evidence is a multiplier on odds. A likelihood ratio above one supports the hypothesis, below one opposes it, and equal to one adds no information.

WHY IT MATTERS IN QUANT RESEARCH

Likelihood ratios make alerts and diagnostic features comparable across populations with different base rates. Log likelihood ratios add across conditionally independent pieces of evidence.

FORMULA INTUITION

Posterior odds = prior odds $\times P(E|H)/P(E|\text{not } H)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

With prior event probability 5%, prior odds are 1:19. Evidence with likelihood ratio 6 produces odds 6:19 and posterior probability $6/25=24\%$, a large update that still falls far short of certainty.

PYTHON / SYSTEM IMPLEMENTATION

Estimate sensitivity and false-positive rate on point-in-time validation data, then compute the ratio with uncertainty. Cap or regularize extreme ratios from sparse cells.

CONFUSIONS TO AVOID

Multiplying likelihood ratios assumes conditional independence when several signals are combined. Reusing correlated evidence counts the same information more than once.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The alert scorer stores each feature's log-likelihood contribution and correlation controls. Explanations show how prior odds became posterior odds step by step.

RELATED CONCEPTS AND QUICK RECALL

Related: diagnostic testing, log odds, evidence weight. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Likelihood is not parameter probability

INTERMEDIATE

ONE-SENTENCE MEANING

A likelihood treats observed data as fixed and compares how parameter values would have generated those data. It need not integrate to one over the parameter and cannot be interpreted as a posterior without a prior and normalization.

MEMORY JOGGER

Probability predicts data from a fixed model; likelihood scores models using fixed data. The same mathematical expression has a different role depending on what varies.

WHY IT MATTERS IN QUANT RESEARCH

Maximum-likelihood estimation, Bayesian updating, and model comparison all use likelihoods. Confusing them with parameter probabilities leads to unjustified certainty and incorrect intervals.

FORMULA INTUITION

$L(\theta; x) = p(x|\theta)$, viewed as a function of θ for fixed x . Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For 7 wins in 10 independent trials, $L(p)$ is proportional to $p^7(1-p)^3$. It ranks candidate win probabilities but does not say there is a 70% probability that p equals 0.7.

PYTHON / SYSTEM IMPLEMENTATION

Implement log-likelihoods to avoid numerical underflow, preserve constants when comparing non-nested models, and test against known toy cases. Record the observation model and independence assumptions.

CONFUSIONS TO AVOID

Normalizing a likelihood curve without a declared prior silently imposes a prior measure. Comparing raw likelihoods across different datasets or sample sizes is not meaningful.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model library exposes separate methods for log-likelihood, fit, posterior, and prediction. The registry records which inferential object each reported interval represents.

RELATED CONCEPTS AND QUICK RECALL

Related: sampling model, MLE, posterior density. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Priors as explicit starting information

INTERMEDIATE

ONE-SENTENCE MEANING

A prior distribution represents uncertainty about a parameter or hypothesis before the current data are incorporated. It may encode historical evidence, structural constraints, or deliberate regularization.

MEMORY JOGGER

A prior is a starting position with a strength, not a guess to hide. Ask what observations it is equivalent to and how conclusions change under credible alternatives.

WHY IT MATTERS IN QUANT RESEARCH

Financial samples are noisy and regimes shift, so priors can stabilize estimates and prevent extreme decisions from small datasets. They also make assumptions auditable rather than implicit in model architecture.

FORMULA INTUITION

Posterior is proportional to likelihood $\times$ prior. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A $\text{Beta}(8,8)$ prior for a hit rate centers at 50% with 16 pseudo-observations. After 3 wins in 4 trades, the posterior is $\text{Beta}(11,9)$, whose mean is 55%, not the raw 75%.

PYTHON / SYSTEM IMPLEMENTATION

Store prior family, parameters, rationale, effective sample size, owner, and approval date. Run prior-predictive checks and sensitivity analysis before examining the final result.

CONFUSIONS TO AVOID

A so-called noninformative prior can still be informative after reparameterization. Choosing a prior after seeing the desired answer is data reuse, not transparent Bayesian analysis.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model registry versions priors separately from code and data. Governance requires a challenger prior and escalation when posterior decisions are prior-sensitive.

RELATED CONCEPTS AND QUICK RECALL

Related: regularization, pseudo-counts, prior predictive. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Posterior distributions and uncertainty

INTERMEDIATE

ONE-SENTENCE MEANING

The posterior distribution combines prior information and observed-data likelihood into updated uncertainty about parameters or hypotheses. Decisions should use its spread and shape, not only its mean or mode.

MEMORY JOGGER

The posterior is what remains plausible after seeing the evidence under the model. More data usually concentrate it, but misspecification can produce confident wrong answers.

WHY IT MATTERS IN QUANT RESEARCH

Posterior probabilities support sizing, model averaging, credible intervals, and probability-of-edge statements. They make estimation uncertainty available to downstream decisions.

FORMULA INTUITION

$p(\theta|x) = p(x|\theta)p(\theta) / \int p(x|\theta)p(\theta)d\theta$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A posterior for expected strategy return may place 72% probability above zero but only 28% above the 4 bp cost hurdle. The second probability is more relevant to deployment.

PYTHON / SYSTEM IMPLEMENTATION

Retain posterior draws or a faithful approximation, compute decision-relevant functionals, and diagnose sampling convergence. Refit only with data available by the declared cutoff.

CONFUSIONS TO AVOID

A 95% credible interval has a posterior interpretation conditional on the model and prior; it is not a guarantee about repeated procedures. Narrow intervals do not protect against omitted regimes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The inference layer publishes posterior draws and summaries to the decision service. The audit record links every draw set to data, code, prior, and random seed.

RELATED CONCEPTS AND QUICK RECALL

Related: credible interval, posterior mean, Bayesian decision theory. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Posterior predictive distributions

INTERMEDIATE

ONE-SENTENCE MEANING

A posterior predictive distribution averages future outcomes over posterior parameter uncertainty. It is wider and more decision-relevant than a prediction that plugs in a single fitted parameter.

MEMORY JOGGER

Predict the next observation under every plausible parameter value, then average. Parameter uncertainty and outcome noise are both part of the forecast.

WHY IT MATTERS IN QUANT RESEARCH

Backtest outcomes, fills, losses, and regime paths should be evaluated through predictive distributions. Predictive checks reveal when a fitted model cannot reproduce important properties of observed data.

FORMULA INTUITION

$p(x_{\text{new}}|x) = \int p(x_{\text{new}}|\theta)p(\theta|x)d\theta$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If a strategy hit-rate posterior is $\text{Beta}(11,9)$, the predictive probability of a win on the next comparable trade is $11/20=55\%$. A multi-trade predictive distribution also includes uncertainty about the unknown hit rate.

PYTHON / SYSTEM IMPLEMENTATION

Simulate θ from the posterior and outcomes from the observation model, preserving dependence and horizon. Compare replicated statistics with observed statistics chosen before fitting.

CONFUSIONS TO AVOID

Posterior fit to parameters does not imply predictive adequacy. Generating independent future trades from a model that ignores serial dependence will understate drawdown uncertainty.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The simulation service consumes posterior artifacts and produces scenario paths for risk and capacity tests. Monitoring compares realized coverage with predictive intervals.

RELATED CONCEPTS AND QUICK RECALL

Related: predictive checks, parameter uncertainty, simulation. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Beta-binomial updating for event rates

INTERMEDIATE

ONE-SENTENCE MEANING

A beta prior combined with binomial data yields a beta posterior by adding successes and failures to prior pseudo-counts. This conjugate model gives a transparent baseline for uncertain event probabilities.

MEMORY JOGGER

Start with alpha successes and beta failures, then add observed counts. The posterior mean shrinks the raw rate toward the prior according to relative sample strength.

WHY IT MATTERS IN QUANT RESEARCH

Hit rates, fill rates, breach rates, and conversion probabilities often begin with this model. It is fast, interpretable, and useful for small-sample stabilization.

FORMULA INTUITION

Prior $\text{Beta}(a,b)$; data s successes and f failures; posterior $\text{Beta}(a+s,b+f)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A $\text{Beta}(2,8)$ prior has mean 20%. Observing 6 successes and 4 failures produces $\text{Beta}(8,12)$, posterior mean 40%, between the 60% sample rate and 20% prior.

PYTHON / SYSTEM IMPLEMENTATION

Use exact beta quantiles for credible intervals and log the effective prior count $a+b$. Extend to beta-binomial or hierarchical models when rates vary across groups.

CONFUSIONS TO AVOID

The basic binomial model assumes exchangeable independent trials with one stable probability. Clustering, changing opportunity quality, and censoring can make the posterior too narrow.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metric service uses beta-binomial baselines for low-count cohorts and exposes raw counts, posterior summaries, and prior sensitivity together.

RELATED CONCEPTS AND QUICK RECALL

Related: conjugacy, shrinkage, exchangeability. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Gaussian updating and precision weighting

INTERMEDIATE

ONE-SENTENCE MEANING

With a normal prior and normal observation model of known variance, the posterior mean is a precision-weighted average of prior mean and sample mean. More precise information receives more weight.

MEMORY JOGGER

Precision is inverse variance, so confidence acts like weight. A noisy sample should move the prior less than a precise sample with the same average.

WHY IT MATTERS IN QUANT RESEARCH

Gaussian updating is a useful approximation for latent returns, factor premia, and sensor-like market estimates. It clarifies how sample size and noise determine shrinkage.

FORMULA INTUITION

posterior precision = prior precision + data precision; posterior mean = precision-weighted means. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A prior mean of 2 bps with standard error 4 bps and a sample mean of 8 bps with standard error 3 bps yield posterior mean about 5.84 bps, closer to the more precise sample.

PYTHON / SYSTEM IMPLEMENTATION

Calculate in precision space, verify units and independence, and expose each information source's weight. Use matrix formulas when parameters are correlated.

CONFUSIONS TO AVOID

Known variance and normality are strong assumptions, especially for heavy-tailed returns. Treating an estimated standard error as fixed can overstate posterior precision.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The signal combiner treats research estimates as noisy measurements and produces shrunk posterior forecasts. The risk layer consumes the full posterior covariance.

RELATED CONCEPTS AND QUICK RECALL

Related: precision, shrinkage, Kalman filtering. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Naive Bayes and evidence dependence

INTERMEDIATE

ONE-SENTENCE MEANING

Naive Bayes multiplies feature likelihoods under a conditional-independence assumption given the class. It can classify well even when the assumption is imperfect, but its probabilities may be poorly calibrated.

MEMORY JOGGER

Each feature casts an evidence vote, and the votes multiply. Correlated votes can make the combined confidence much stronger than the information warrants.

WHY IT MATTERS IN QUANT RESEARCH

Fast text-like event classifiers, market-state labels, and alert triage can use naive Bayes as an interpretable baseline. Its feature contributions are easy to audit.

FORMULA INTUITION

$P(H|x_1, \dots, x_k)$ proportional to $P(H)$ product $_j P(x_j|H)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Two volatility indicators derived from the same returns both favor stress. Multiplying them as independent evidence effectively counts one volatility shock twice and can inflate posterior stress probability.

PYTHON / SYSTEM IMPLEMENTATION

Estimate class-conditional feature models with smoothing, inspect residual dependence, and recalibrate probabilities on untouched data. Group or remove near-duplicate features.

CONFUSIONS TO AVOID

Good ranking performance does not imply trustworthy probabilities. Zero-frequency categories require smoothing, and distribution drift can invalidate likelihood estimates rapidly.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The baseline model registry includes feature-level log-evidence, dependency diagnostics, and a calibrated output layer. Drift monitors watch both class prevalence and feature likelihoods.

RELATED CONCEPTS AND QUICK RECALL

Related: conditional independence, calibration, generative classifier. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Sequential Bayesian updating

INTERMEDIATE

ONE-SENTENCE MEANING

Sequential updating uses one posterior as the next prior when observations arrive in order. Under a stable model, processing data one batch at a time yields the same posterior as processing all data together.

MEMORY JOGGER

Learn, carry the belief forward, then learn again. The update order should not matter mathematically, but timestamping and model changes matter operationally.

WHY IT MATTERS IN QUANT RESEARCH

Online regime inference, execution quality, and strategy health monitoring require continuous belief updates. Sequential structure supports real-time decisions while retaining uncertainty.

FORMULA INTUITION

$p(\theta|x_{1:t})$ proportional to $p(x_t|\theta)p(\theta|x_{1:t-1})$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Starting from $\text{Beta}(2,2)$, 7 wins and 3 losses produce $\text{Beta}(9,5)$. A later batch of 2 wins and 8 losses gives $\text{Beta}(11,13)$, exactly matching one update with 9 wins and 11 losses.

PYTHON / SYSTEM IMPLEMENTATION

Persist posterior state atomically with the last processed event ID, support replay, and make updates idempotent. Version observation models so a model change triggers a controlled reset or migration.

CONFUSIONS TO AVOID

Reprocessing events double-counts evidence. If the environment changes, endlessly accumulating old data can make the posterior slow to adapt and falsely precise.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The streaming inference service checkpoints posterior state and data offsets. A shadow batch calculation reconciles online and offline results on a schedule.

RELATED CONCEPTS AND QUICK RECALL

Related: online learning, state-space models, forgetting. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Bayesian regime probabilities

INTERMEDIATE

ONE-SENTENCE MEANING

A regime model assigns posterior probabilities to latent market states rather than forcing a single hard label. Forecasts can be averaged across states according to those probabilities.

MEMORY JOGGER

Markets rarely announce a regime boundary. Carry uncertainty over states instead of pretending the most likely state is certain.

WHY IT MATTERS IN QUANT RESEARCH

Soft regime weights reduce discontinuous strategy switching and support scenario-aware risk. They also expose ambiguity when several states explain current observations similarly.

FORMULA INTUITION

$P(S_t | \text{data}) \propto P(\text{observation}_t | S_t) \sum_j P(S_t | S_{t-1} = j) P(S_{t-1} = j | \text{past})$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

After a volatility jump, posterior state weights may be 60% stress, 30% transition, and 10% calm. A risk forecast can mix state-specific covariances using those weights.

PYTHON / SYSTEM IMPLEMENTATION

Use forward filtering with log probabilities, stable transition matrices, and point-in-time features. Validate state interpretation, dwell times, and forecast value out of sample.

CONFUSIONS TO AVOID

States can swap labels across fits and may reflect statistical artifacts rather than economic regimes. Hard-coding strategy behavior to unstable labels creates brittle controls.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The regime service emits posterior weights and uncertainty; allocation blends approved state policies and falls back when likelihoods are out of support.

RELATED CONCEPTS AND QUICK RECALL

Related: hidden Markov models, filtering, mixture forecasts. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Signal precision, recall, and base-rate shift

INTERMEDIATE

ONE-SENTENCE MEANING

Precision is $P(\text{outcome}|\text{signal})$, while recall is $P(\text{signal}|\text{outcome})$; both depend on event definitions, and precision additionally depends strongly on outcome prevalence. A model can retain sensitivity while losing economic usefulness after a base-rate shift.

MEMORY JOGGER

Recall asks how many events were caught; precision asks how many alerts were right. Decision capacity and cost usually make precision operationally critical.

WHY IT MATTERS IN QUANT RESEARCH

Trade selection, risk alerts, and anomaly detectors operate under asymmetric costs. Thresholds should optimize expected utility rather than a prevalence-insensitive score alone.

FORMULA INTUITION

$\text{precision} = \text{TP}/(\text{TP}+\text{FP})$; $\text{recall} = \text{TP}/(\text{TP}+\text{FN})$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

An alert with 80% recall and 10% false-positive rate has 47% precision when event prevalence is 10%, but only 7.5% precision when prevalence falls to 1%.

PYTHON / SYSTEM IMPLEMENTATION

Monitor prevalence, confusion counts, calibration, and utility at the deployment threshold. Recalibrate or adjust odds when label-conditional behavior is stable but base rates shift.

CONFUSIONS TO AVOID

Accuracy can be high by predicting the majority class. Comparing precision across periods without reporting prevalence can confuse population change with model deterioration.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The alert platform publishes threshold-level decision tables and cost-weighted utility. Drift monitors distinguish prior shift from changes in class-conditional feature behavior.

RELATED CONCEPTS AND QUICK RECALL

Related: confusion matrix, prior probability shift, decision threshold. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Event studies as conditional estimates

INTERMEDIATE

ONE-SENTENCE MEANING

An event study estimates outcomes conditional on a defined event and compares them with an appropriate counterfactual or benchmark. The event definition, overlap, and selection process determine what the estimate means.

MEMORY JOGGER

An event-conditioned average is not automatically a causal effect. Ask what would have happened to comparable observations without the event.

WHY IT MATTERS IN QUANT RESEARCH

Earnings, breakouts, volatility shocks, and macro releases are common research conditions. Proper controls separate market-wide movement, anticipation, and event clustering from event-specific behavior.

FORMULA INTUITION

$\text{abnormal return} = \text{observed return} - \text{expected benchmark return}$; aggregate over declared window. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A post-earnings drift estimate conditions on a surprise known at announcement time, aligns returns to that timestamp, and compares with a factor-adjusted benchmark over non-overlapping windows.

PYTHON / SYSTEM IMPLEMENTATION

Maintain an immutable event calendar with announcement and availability timestamps, deduplicate revisions, define pre/post windows, and cluster uncertainty by issuer and date.

CONFUSIONS TO AVOID

Using scheduled dates instead of actual timestamps can misalign information. Overlapping events, survivorship, and testing many windows can create false conditional patterns.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The event store joins point-in-time events to market data; the study engine produces cohort counts, abnormal-return paths, uncertainty, and multiple-testing metadata.

RELATED CONCEPTS AND QUICK RECALL

Related: counterfactual, abnormal return, event time. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Multiple testing and posterior false discoveries

INTERMEDIATE

ONE-SENTENCE MEANING

Searching many signals raises the chance that an apparently strong conditional pattern is noise. A posterior probability is not protected from this problem if the prior or model space ignores the search process.

MEMORY JOGGER

The more doors opened, the less surprising one lucky result becomes. Count the research paths, not only the final reported test.

WHY IT MATTERS IN QUANT RESEARCH

Quant research explores features, assets, horizons, and thresholds. Selection-aware inference and holdout validation reduce the probability of deploying a false discovery.

FORMULA INTUITION

posterior odds for a selected hypothesis must reflect prior odds across the searched model space. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If 1,000 independent null signals are screened, several can show impressive backtests by chance. Treating the winner as if it were the only planned hypothesis produces overconfident evidence.

PYTHON / SYSTEM IMPLEMENTATION

Log all trials, use hierarchical shrinkage, reserve untouched data, and require replication across regimes or markets. Model the selection step when reporting posterior claims.

CONFUSIONS TO AVOID

A Bayes factor or posterior computed after cherry-picking a specification can still be misleading. Reusing the holdout to refine the model turns it into training data.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment tracker records every submitted configuration and links promoted models to their search family. Governance budgets repeated access to protected evaluation sets.

RELATED CONCEPTS AND QUICK RECALL

Related: false discovery rate, selection bias, research ledger. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Calibration and reliability

INTERMEDIATE

ONE-SENTENCE MEANING

A probability forecast is calibrated when events assigned probability p occur about p fraction of the time across comparable cases. Calibration evaluates probability meaning, not merely ranking ability.

MEMORY JOGGER

Among forecasts near 70%, roughly seven in ten should occur. Check this across time, cohorts, and confidence ranges rather than trusting one pooled curve.

WHY IT MATTERS IN QUANT RESEARCH

Sizing, risk limits, and expected-value decisions require calibrated probabilities. A well-ranked but overconfident model can systematically overallocate capital.

FORMULA INTUITION

Brier score = $\text{mean}((\text{forecast probability} - \text{outcome})^2)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If 100 trades receive forecasts near 80% but only 60 win, the model is overconfident in that range. A ranking metric may still look good if higher scores generally win more often.

PYTHON / SYSTEM IMPLEMENTATION

Use out-of-sample reliability bins with counts and intervals, isotonic or logistic recalibration, and time-sliced checks. Preserve the raw score and calibrator versions separately.

CONFUSIONS TO AVOID

Small bins create noisy curves, while broad bins hide local failures. Recalibrating on the same sample used for evaluation overstates improvement.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The probability service publishes calibration curves, Brier decomposition, and cohort alerts. Downstream decisions reject forecasts from stale or unsupported calibrators.

RELATED CONCEPTS AND QUICK RECALL

Related: reliability diagram, Brier score, discrimination. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Posterior predictive checks and model criticism

INTERMEDIATE

ONE-SENTENCE MEANING

Posterior predictive checks compare observed data with replicated data generated from the fitted model. They test whether the model can reproduce decision-relevant patterns, not whether every discrepancy is statistically significant.

MEMORY JOGGER

Fit the model, simulate worlds from it, and ask whether the real world looks plausible among them. Choose diagnostics tied to the model's intended use.

WHY IT MATTERS IN QUANT RESEARCH

Return models should reproduce tails, volatility clustering, drawdowns, cross-sectional dispersion, and event frequencies relevant to allocation and risk. A model that fits means but misses paths can be dangerous.

FORMULA INTUITION

replicate θ from posterior, then y_{rep} from $p(y|\theta)$, and compare $T(y_{\text{rep}})$ with $T(y)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A return model matches daily mean and variance but generates maximum drawdowns far smaller than observed. The predictive check identifies missing dependence or regime structure.

PYTHON / SYSTEM IMPLEMENTATION

Predefine discrepancy statistics, simulate enough replicates, and visualize observed values within predictive distributions. Use held-out predictive checks for generalization claims.

CONFUSIONS TO AVOID

A posterior predictive p-value is not an ordinary frequentist p-value. Passing chosen checks does not prove the model; it only means those discrepancies were not exposed.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The validation pipeline stores predictive simulations and diagnostic summaries alongside model artifacts. Failed critical checks block promotion or require a documented risk acceptance.

RELATED CONCEPTS AND QUICK RECALL

Related: model criticism, simulation, predictive adequacy. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Bayesian model comparison and averaging

INTERMEDIATE

ONE-SENTENCE MEANING

Bayesian model comparison weighs models by prior plausibility and their integrated ability to predict observed data. Model averaging carries structural uncertainty into forecasts instead of selecting one winner with certainty.

MEMORY JOGGER

A flexible model must earn its complexity by predicting data across its prior parameter range. Average forecasts when several models remain plausible.

WHY IT MATTERS IN QUANT RESEARCH

Alternative factor, volatility, and regime models can produce different decisions. Averaging reduces selection risk when no model dominates robustly.

FORMULA INTUITION

$P(M_k|D)$ proportional to $P(D|M_k)P(M_k)$; predictive = $\sum_k P(y|M_k,D)P(M_k|D)$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If posterior model weights are 0.5, 0.3, and 0.2 for three return models, the combined forecast is their weighted predictive mixture, including between-model dispersion.

PYTHON / SYSTEM IMPLEMENTATION

Specify model priors before fitting, compute marginal likelihoods with validated methods, and compare predictive performance. Store model-specific and averaged forecasts for attribution.

CONFUSIONS TO AVOID

Marginal likelihoods can be extremely prior-sensitive. Selecting models by in-sample fit and then treating the winner as certain discards structural uncertainty.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The ensemble service consumes approved model artifacts and versioned weights. Monitoring tracks weight concentration, disagreement, and fallback behavior.

RELATED CONCEPTS AND QUICK RECALL

Related: Bayes factor, marginal likelihood, ensemble. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Decision thresholds and expected utility

INTERMEDIATE

ONE-SENTENCE MEANING

A posterior probability becomes actionable only after consequences are mapped into gains, losses, constraints, and alternatives. The optimal threshold depends on payoff asymmetry, not on probability alone.

MEMORY JOGGER

Probability describes belief; utility chooses action. A 60% chance can justify or reject a trade depending on upside, downside, costs, and opportunity cost.

WHY IT MATTERS IN QUANT RESEARCH

Trading, risk alerts, and model promotion all require explicit decision loss. Utility-based thresholds align probability forecasts with capital outcomes.

FORMULA INTUITION

choose action a maximizing $E[U(a, \theta) | \text{data}]$. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A trade wins 55% of the time, gains 1R on a win, loses 1R on a loss, and costs 0.15R. Expected value is $0.55 - 0.45 - 0.15 = -0.05R$, so the probability edge is insufficient.

PYTHON / SYSTEM IMPLEMENTATION

Version payoff tables, constraints, and thresholds separately from the probability model. Evaluate realized utility and regret, not only classification metrics.

CONFUSIONS TO AVOID

Optimizing a threshold on the same backtest overfits both model and policy. Utility inputs can be state-dependent and uncertain, so sensitivity analysis is required.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The decision engine combines posterior forecasts with a governed payoff model and risk constraints. Audit logs retain the rejected alternatives and expected utility at decision time.

RELATED CONCEPTS AND QUICK RECALL

Related: Bayesian decision theory, cost-sensitive classification, threshold. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Production lineage, monitoring, and rollback

INTERMEDIATE

ONE-SENTENCE MEANING

A production probability is a derived artifact whose event definitions, priors, likelihoods, calibration, data cutoff, and code version must be recoverable. Monitoring must detect both data drift and broken probability meaning.

MEMORY JOGGER

A probability without lineage cannot be audited, replayed, or safely changed. Treat definitions and inference state as production data, not notebook annotations.

WHY IT MATTERS IN QUANT RESEARCH

Quant systems make repeated capital decisions, so silent denominator changes or stale priors can accumulate large losses. Operational controls are part of statistical validity.

FORMULA INTUITION

published artifact = estimate + uncertainty + cohort + as-of time + lineage + health state. Verify the denominator, support, timing, and estimation assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A fill model remains numerically available after a venue schema change but begins excluding partial fills. Coverage and calibration drift should trip a health gate before routing decisions use it.

PYTHON / SYSTEM IMPLEMENTATION

Log source hashes, event contracts, posterior state, calibrator, code, dependencies, and decision IDs. Add replay tests, shadow estimates, freshness limits, drift thresholds, and tested rollback artifacts.

CONFUSIONS TO AVOID

Monitoring only average prediction hides cohort failures. Retraining automatically on drift can absorb corrupted labels or temporary shocks and make rollback difficult.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model registry, feature store, inference service, and decision engine share immutable artifact IDs. A control plane can disable, fall back, or roll back a probability source independently.

RELATED CONCEPTS AND QUICK RECALL

Related: model registry, data lineage, observability. State the denominator, name the information available at the cutoff, and identify one diagnostic that could overturn the conclusion.

Probability tree: signal to outcome

INTERMEDIATE

PURPOSE

This tree separates the base rate of a signal from conditional execution and outcome probabilities. Multiply along a path, then add terminal paths that answer the same final question.



WORKED PATH

$P(\text{signal and fill and win}) = 0.10 \times 0.80 \times 0.55 = 4.4\%$ of eligible days. Conditional on a signal, the winning filled-trade probability is $0.80 \times 0.55 = 44\%$; the denominator determines which number answers the decision.

AUDIT QUESTIONS

Do all child branches sum to one? Are fill and outcome rates estimated on their parent populations? Was signal availability known before fill and return? Are non-fills excluded from P&L intentionally or hiding opportunity cost?

SYSTEM CONNECTION

Serialize nodes and edges with counts, uncertainty, and as-of timestamps. The same tree can reconcile research forecasts, live funnel metrics, and scenario stress by replacing selected branch probabilities.

Bayesian update: rare-event alert

INTERMEDIATE

QUESTION

An event occurs on 1% of days. An alert catches 80% of event days and fires falsely on 10% of ordinary days; what is $P(\text{event}|\text{alert})$?

Prior event



1.00%

True-alert mass



0.80%

False-alert mass



9.90%

Posterior event



7.48%

CALCULATION

True-alert mass = $0.01 \times 0.80 = 0.008$. False-alert mass = $0.99 \times 0.10 = 0.099$. Total alert mass is 0.107, so $P(\text{event}|\text{alert}) = 0.008/0.107 = 7.48\%$.

INTERPRETATION

The alert is informative: it raises event probability from 1% to 7.48%, a 7.5-fold increase. It is not definitive because false alerts from the much larger ordinary population outnumber true alerts by more than twelve to one.

DECISION CONNECTION

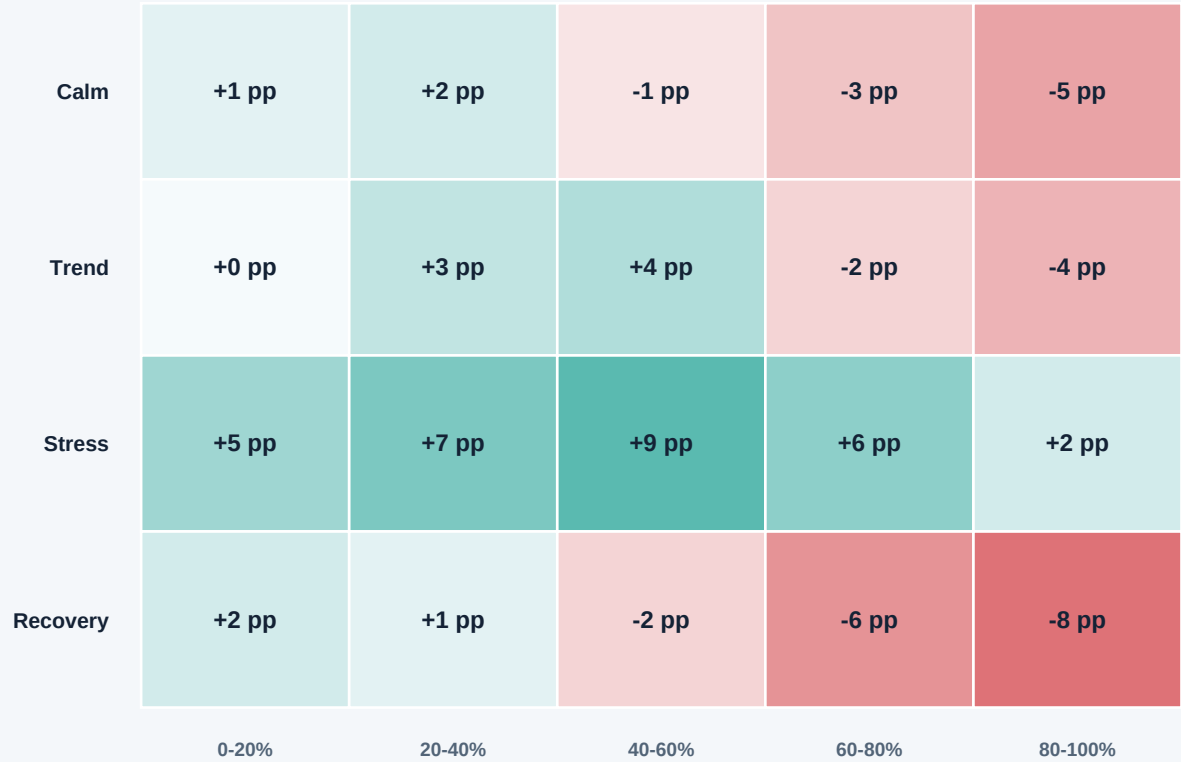
Whether 7.48% warrants action depends on the loss prevented, action cost, and capacity. Monitor prior prevalence, sensitivity, and false-positive rate separately because each can drift and change the posterior.

Calibration across regimes

INTERMEDIATE

READING RULE

Each cell shows observed frequency minus predicted probability in percentage points. Positive values mean underprediction; negative values mean overprediction. Counts and intervals are required before escalating a cell.



PATTERN

Stress rows are positive across every forecast range, so realized event rates exceed probabilities throughout that regime. Recovery shows the opposite in high-score cells, suggesting stale stress evidence or a calibrator that adapts too slowly.

INVESTIGATION SEQUENCE

Verify labels and regime timing, add counts and confidence intervals, compare raw scores with calibrated probabilities, then inspect prevalence and feature-likelihood drift. Recalibrate only on a new training window and preserve an untouched evaluation period.

PRODUCTION GUARDRAIL

Global calibration can look acceptable while a capital-intensive cohort is badly wrong. Health gates should combine pooled and material-cohort thresholds, with a fallback probability source and tested rollback.

Formula sheet and identity map

INTERMEDIATE

CONDITIONING

$P(A|B) = P(A \text{ and } B) / P(B)$. The denominator is the probability mass of the conditioning event, so the formula is undefined when $P(B) = 0$ and unstable when the observed B sample is tiny.

MULTIPLICATION

$P(A \text{ and } B) = P(A|B)P(B) = P(B|A)P(A)$. Extend it sequentially to factor any joint distribution into conditional terms.

TOTAL PROBABILITY

For a complete, disjoint partition $\{B_i\}$, $P(A) = \sum_i P(A|B_i)P(B_i)$. Cell contributions reveal whether the aggregate is changing because of state mix or within-state behavior.

BAYES

$P(H|E) = P(E|H)P(H) / \sum_j P(E|H_j)P(H_j)$. The denominator normalizes unscaled hypothesis weights so posterior mass sums to one.

ODDS UPDATE

Posterior odds equal prior odds times the likelihood ratio. In log space, independent evidence contributions add, which improves numerical stability and interpretability.

PREDICTION

$p(x_{\text{new}}|x) = \int p(x_{\text{new}}|\theta)p(\theta|x)d\theta$. Posterior predictive uncertainty includes both random outcome noise and uncertainty about θ .

Worked calculations and interpretation

INTERMEDIATE

REVERSE CONDITIONING

Out of 1,000 days, 10 are crashes. An alert fires on 8 crash days and 99 ordinary days. $P(\text{crash}|\text{alert})=8/107=7.48\%$, while $P(\text{alert}|\text{crash})=8/10=80\%$; the shared numerator does not make the rates interchangeable.

TOTAL PROBABILITY

A signal wins 60% in a 30%-weight trend regime and 45% otherwise. Its marginal win rate is $0.30 \times 0.60 + 0.70 \times 0.45=49.5\%$.

LIKELIHOOD RATIO

A feature occurs on 50% of stress days and 10% of calm days, so $LR=5$. With prior stress probability 20%, prior odds 1:4 become posterior odds 5:4, or 55.6%.

BETA-BINOMIAL

A $\text{Beta}(4,6)$ prior updated with 12 wins and 8 losses becomes $\text{Beta}(16,14)$. The posterior mean is 53.3%, between the 40% prior mean and 60% sample rate.

PRECISION WEIGHTING

A 0 bp prior estimate with standard error 4 and a 6 bp sample estimate with standard error 2 produce weights proportional to $1/16$ and $1/4$. The posterior mean is 4.8 bps.

DECISION THRESHOLD

A protective action costs 1 unit and avoids 20 units when a loss event occurs. Ignoring other effects, act when posterior event probability exceeds $1/20=5\%$.

Failure modes and diagnostic table

INTERMEDIATE

POSTERIOR IS IMPLAUSIBLY HIGH

Check whether $P(E|H)$ was mistaken for $P(H|E)$, whether the base rate was omitted, and whether correlated evidence was multiplied as independent.

CONDITIONAL RATE JUMPS

Inspect raw counts, denominator policy, population mix, event timing, and missing or censored records. Attach intervals before labeling the movement structural.

AGGREGATE AND SUBGROUP STORIES CONFLICT

Recompute common-weight standardized rates, inspect group-size shifts, and determine whether the grouping variable is a confounder, mediator, or collider.

CALIBRATION DECAYS

Separate prevalence shift from class-conditional drift, verify labels, compare score and probability distributions, and evaluate material cohorts with enough count.

POSTERIOR BECOMES TOO NARROW

Test independence, exchangeability, likelihood tails, prior strength, autocorrelation, and repeated evidence. Use clustered or hierarchical structure where appropriate.

ONLINE AND BATCH RESULTS DIFFER

Check duplicate events, ordering, late data, state checkpoints, model-version changes, numerical underflow, and whether the update is idempotent.

Implementation and model-registry checklist

INTERMEDIATE

EVENT CONTRACT

Observation grain, entity key, population, event expression, boundaries, horizon, decision cutoff, availability timestamp, missing policy, and owner.

PROBABILITY CONTRACT

Conditional direction, numerator, denominator, weights, sample count, effective count, interval method, support, fallback, and refresh cadence.

BAYESIAN CONTRACT

Hypothesis space, prior family and strength, likelihood, conditional-independence assumptions, normalization, inference method, seed, and diagnostics.

POINT-IN-TIME CONTROLS

Immutable source hashes, feature lags, historical universes, revision handling, sequential offsets, replay test, and future-perturbation leakage test.

VALIDATION EVIDENCE

Confusion counts, reliability curves, Brier score, regime slices, prior sensitivity, posterior predictive checks, decision utility, and protected holdout results.

RUNTIME GOVERNANCE

Artifact ID, code and dependency versions, health state, freshness, drift limits, alert playbook, shadow estimate, fallback, approval, and rollback.

Final active recall index

INTERMEDIATE

FOUNDATIONS

Define sample space, event, joint, marginal, conditional, complement, union, independence, partition, total probability, and base rate without changing the observation grain.

CONDITIONING

Build a two-way table and tree; explain conditional distributions, continuous conditions, conditional mean and variance, iterated expectation, selection bias, and Simpson's paradox.

BAYES

Derive the normalized and odds forms, distinguish likelihood from posterior, defend a prior, compute beta-binomial and Gaussian updates, and explain posterior prediction.

MODEL DIAGNOSTICS

Audit support, dependence, prior sensitivity, selection, multiple testing, calibration, predictive adequacy, and model disagreement before trusting a probability.

DECISION USE

Translate posterior beliefs into expected utility with explicit gains, losses, costs, capacity, constraints, alternatives, and uncertainty in the payoff model.

PROMOTION TEST

Before publishing, answer: conditional on what, relative to which base rate, known when, estimated how, how uncertain, calibrated where, monitored how, and consumed by which decision?